Grupo N° 8.

II.d.1.1. Expresar en forma decimal.

a) $10/20 = 0,75$

Calculos

$$\frac{15}{20} = \frac{15/5}{20/5} = \frac{3}{4} = \boxed{0,75}$$

II.d.1.2 Expresar en forma racional.

a) $1,222...$

Calculos

$x = 1,222$

$10x = 1,222...$

$10x - x = 12,222... - 1,222...$

$9x = 11$

$x = \frac{11}{9}$

$$\boxed{1,2222... = \frac{11}{9}}$$

II.d.1.3 Resolver los siguientes calculos combinados y expresar el resultado como número racional en su mínima expresión.

c) $$\frac{\left(0,\overline{54} + \frac{3}{5}\right)^2}{0,10\overline{8}}$$

$$\frac{\left(\frac{49}{90} + \frac{3}{5}\right)^2}{\frac{49}{450}} =$$

Calculos

$x = 0,5444...$

$100x = 54,444...$

$100x - 10x = 54,\cancel{444}... - 5,\cancel{444}...$

$90x = 49$

$$\boxed{x = \frac{49}{90}}$$

$x = 0,10\overline{8}$

$100x = 10,888...$

$1000x - 100x = 108,\cancel{888}... - 10,\cancel{888}...$

$900x = 98$

$x = \frac{98}{900}$

$$\boxed{x = \frac{49}{450}}$$

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$$\frac{\left(\frac{49}{90}+\frac{3}{5}\right)^2}{\frac{49}{450}} =$$

$$\frac{\left(\frac{49+3\cdot 18}{90}\right)}{\frac{49}{450}} =$$

$$\left(\frac{103}{90}\right)^2 \cdot \frac{450}{49} =$$

$$\boxed{\frac{10609}{882}}$$

$$\left(\frac{49}{90}+\frac{3}{5}\right)^2 =$$

18 es porque $18 \cdot 5 = 90$

$$\left(\frac{49+3\cdot 18}{90}\right)$$

$$\left(\frac{103}{90}\right)^2 \cdot \frac{450}{49} =$$

simplificamos [4100 : 450 =

$$\frac{10609}{8100} \cdot \frac{450}{49} =$$

$$\frac{10609}{18\cdot 49} = \frac{10609}{882}$$

II.d.1.B Resolver los siguientes cálculos combinados y expresar el resultado como número racional con su mínima expresión.

d) $\sqrt{\dfrac{0.\overline{8}}{(0,00\overline{7})^3}} =$

II. d. 1. 4 Guía de ejercicios con operaciones de números racionales.

e) $-\frac{3}{4} \cdot \left[\frac{4}{3}\left(\frac{1}{2} - \frac{1}{3}\right) + \frac{2}{7}\right]$

$-\frac{3}{4}\left[\frac{4}{3} \cdot \left(\frac{1}{6}\right) + \frac{2}{7}\right]$

$-\frac{3}{4}\left[\frac{2}{9} + \frac{2}{7}\right]$

$-\frac{3}{\cancel{4}} \cdot \frac{\cancel{32}}{63} =$

$-\frac{\cancel{96}}{\cancel{252}} = \boxed{-\frac{8}{21}}$

▷ simplificamos dividiendo por 12.

Calculos

$\frac{1}{2} - \frac{1}{3} = \frac{1}{2} + \left(-\frac{1}{3}\right) = \frac{3-(-2)}{6}$

$\boxed{= \frac{1}{6}}$

$\frac{2}{9} + \frac{2}{7} = \frac{14+18}{63} =$

$\boxed{\frac{32}{63}}$

$\frac{4}{3} \cdot \frac{1}{6} = \frac{4}{18} = \boxed{\frac{2}{9}}$

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g) $8 - \frac{\frac{2}{5} - \frac{1}{9}}{\frac{7}{6}} + \frac{1}{21}\left(\frac{9}{4} - \frac{2}{5}\right)$

$8 - \frac{\frac{2}{5} - \frac{1}{9}}{\frac{7}{9}} + \frac{1}{21} + \frac{37}{20}$

$8 - \frac{\frac{13}{45}}{\frac{7}{6}} + \frac{1}{21} \cdot \frac{37}{20}$

$8 - \frac{26}{105} + \frac{37}{420}$

$8 - \frac{67}{420}$

$\frac{3360}{420} - \frac{67}{420} = \boxed{\frac{3293}{420}}$

Calculos

$\frac{9}{9} - \frac{2}{5} = \frac{45-8}{20} = \boxed{\frac{37}{20}}$

$\frac{\frac{13}{45}}{\frac{7}{6}} = \frac{13}{45} \cdot \frac{6}{7} = \frac{78}{315} = \boxed{\frac{26}{10}}$

$\frac{1}{21} \cdot \frac{37}{20} = \frac{37}{420}$

$8 - \frac{26}{105} + \frac{37}{420}$

$\frac{8 \times 420}{420} = \frac{3360}{420}$

$\frac{-104}{420} + \frac{37}{420} = \boxed{\frac{-67}{420}}$

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